

τ -Rec: A Verifiable Benchmark for Agentic Recommender Systems

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A recommender agent has to ask about needs, check the catalog, and commit to one pick.



The best agent solves 57% of tasks on one try and 35% on all 4 tries.

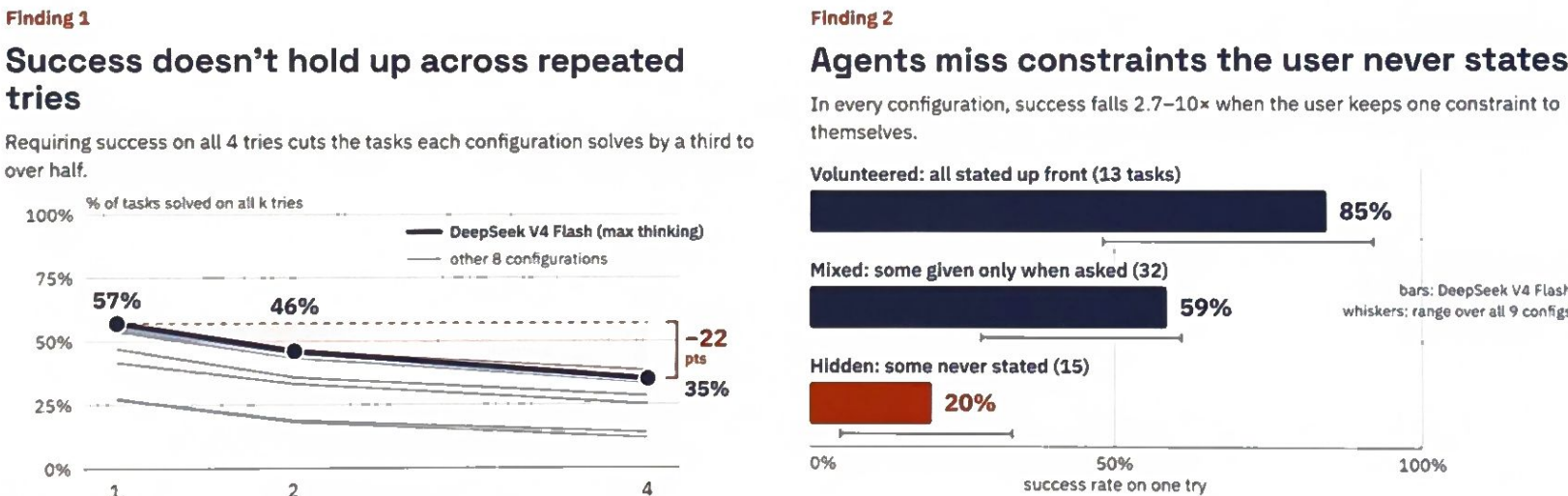
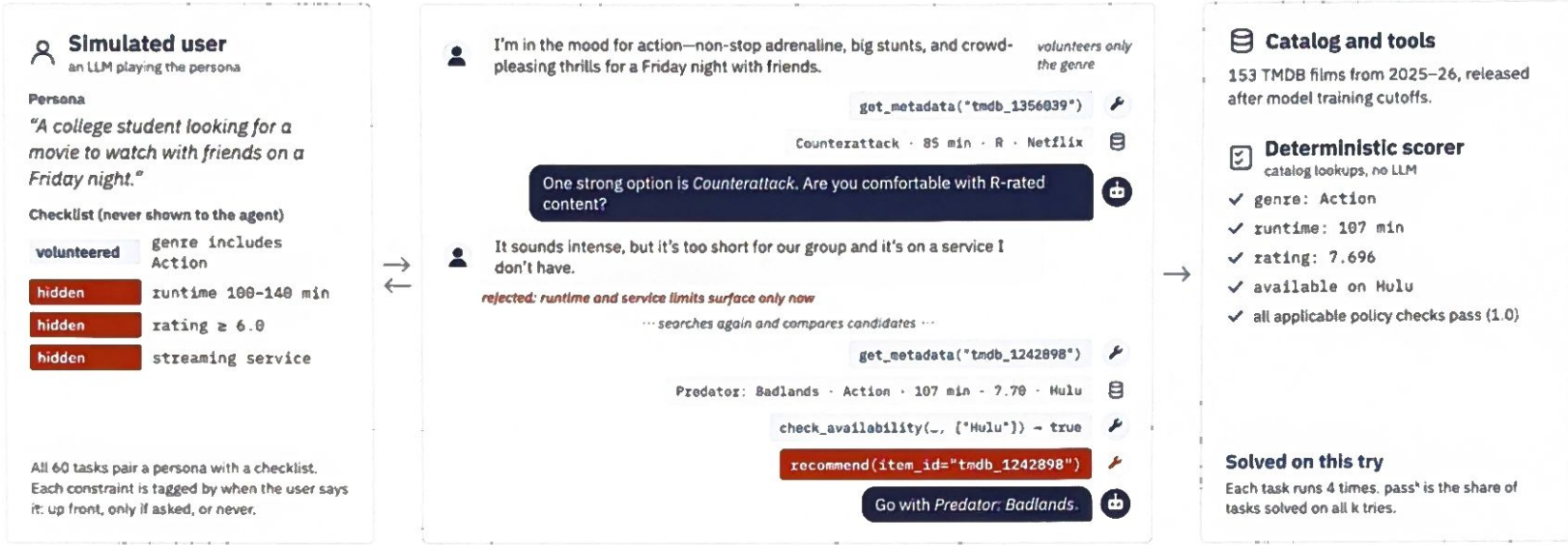


τ -Rec tests this on 60 multi-turn movie tasks and scores each pick against the catalog, without an LLM judge.



Success falls from 85% to 20% when the user keeps one need unstated.

How a τ -Rec task runs



Finding 3

The best agents check before they commit

The hedger

Ends without a recommendation, though only 5 of 60 tasks call for that.

No recommendation issued

Gemini 2.5 Flash	67%
GPT-5 mini	45%
Qwen3-32B	43%

The bluffer

Recommends without checking streaming availability.

Availability violations

Qwen3-32B	21%
Sonnet 4.6	9%
GPT-5.4 (no thinking)	8%

The verifier

Checks the catalog before committing and reaches the top success rates (54-57%).

Median tool calls per task

DeepSeek V4 Flash	28-30
GPT-5.4 + thinking	25
Sonnet 4.6	16

How τ -Rec compares to prior CRS resources

✓ full ○ partial

Builds on τ -bench (Yao et al., 2025) and applies its pass^{*} metric and simulated user to recommendation.

	Verifiable rewards	Multi-turn	Tool use	Policy checks	Fresh catalog	pass [*]	Hidden intent
CRSLab		○					
iEvalM		✓					○
UserSimCRS v2	○	✓					○
AgentRecBench	○		✓				
InteRecAgent		✓	✓				○
MATCHA		✓	✓	○			
τ -Rec (ours)	✓	✓	✓	✓	✓	✓	✓

Get the benchmark

Paper arXiv 2606.10156	Code GitHub	Dataset Hugging Face	Contact nbharaths@gmail.com